

XP Man (Experiment Manager)

Experiment manager is an application that has been developed in Java programming language by

Andrea CONTE
Face Categorization Lab
Institute of Psychology (IPSY)
Institute of Neuroscience (IONS)
University of Louvain

Purpose

Its purpose is first to be used in labs as a testing subject tool whatever task is required.

The application is organised in a way that makes your testing simpler, flexible and well-organised. The testing task must be developed apart in Java language and be inserted in the application as a module. The Fast Periodic Visual Stimulation task (FPVS) module is already integrated in the application by default.

System you can use it on

- Windows XP (recommended)
- Windows 7 (recommended)
- Windows 8 & 10 (not tested a lot)
(Windows 10 first testings seems good)
- Mac OS X 10.x (Would recommend minimum 10.8)

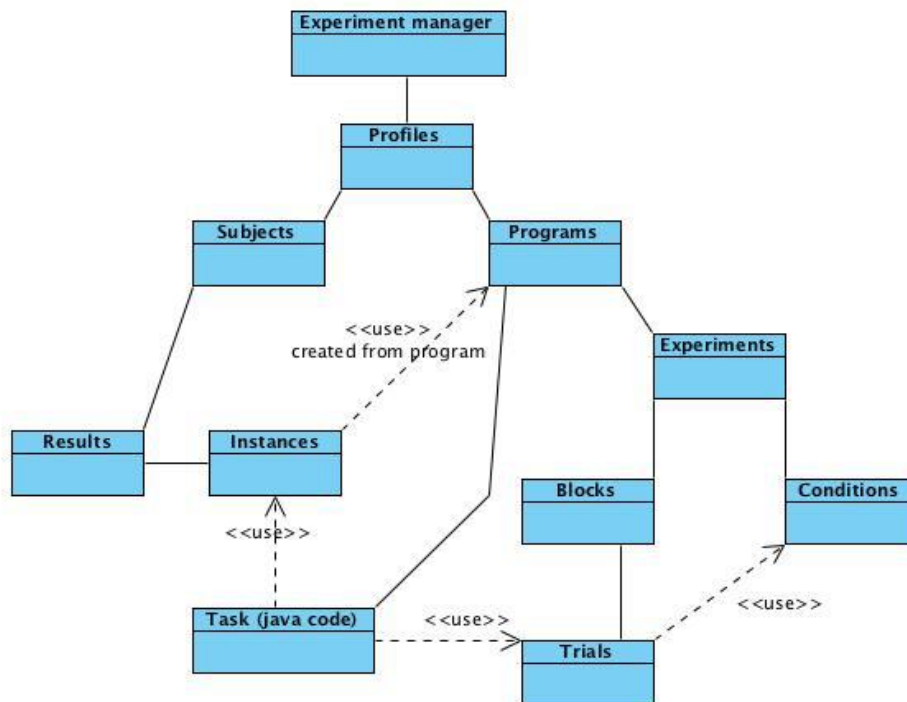
Minimum java version to be installed

For Windows: you need minimum Java version 8 JRE (Java Runtime Edition)

For Mac OS X: you need minimum Java version 8 JDK (Java Development Kit)

(Important to install the JDK version for Mac OS X otherwise it will not work)

Basic concepts



As you can see in the diagram, the application is composed of **Profiles**.

A **profile** is what defines you (the experimenter).

You can think of a profile as a session like in Windows or Mac OSX. You can then either put a password or not to protect that profile. A profile is composed of subjects and programs.

A **subject** is the basic information you need to identify a person, you need to create a subject first to make a person pass a task. When a subject finishes a task, results of that task will be added to this subject result file.

A **program** is the start point of experiment definitions. A program is then composed of experiments. A program can have parameters attached to it (like screen parameters: full screen mode, viewport resolution, etc.) A program defines the task that will be launched later and that was created in Java programming language.

A **task** is the java code that will be launched using the parameters defined in the program, and the experiment parameters and the condition parameters both defined in that program.

An **experiment** is composed of conditions and blocks of trials. An experiment can also have parameters attached to it (like fixation cross parameters, the timing parameters, etc.)

A **trial** is one run of some condition parameters. A trial is then linked to a condition that defined the trial parameters. Trials will be launched following each other when the task starts.

A **block** is composed of trials. It serves to organise trials.

A **condition** defined the parameters that will be used in a trial. A condition is linked to a trial. The same condition can be linked to many trials as well. Condition parameters could be the paradigm type selection, the stimuli directory selection, the stimuli sequence pattern selection, etc.

An **instance** or 'program instance' is a photograph of a program. Once a program instance is created, that instance parameters cannot be changed. If you need to make changes, you have to go back to the program definitions, make the changes, and then create a new instance to include those changes.

How to install it?

First and if not installed already, install the Java Runtime Standard Edition (Java SE)

You will need minimum the Java Runtime SE version 8.

If not installed on your computer, you need to download and install it (see link below).

- For **Windows users**, you will need to download the JRE (Java Runtime Environment)
 - If you have Windows in 32 bits, you must download the Windows x86 version
 - If you have Windows in 64 bits, you must download the Windows x64 version
- For **Mac OSX users**, you will need to download the JDK (Java Development Kit instead of the JRE).
 - If you have Mac OS X, download the Mac OS X x64 version

Important: You have to install the JDK version for Mac OS X otherwise it will not work

Download link: <http://www.oracle.com/technetwork/java/javase/downloads/index.html>

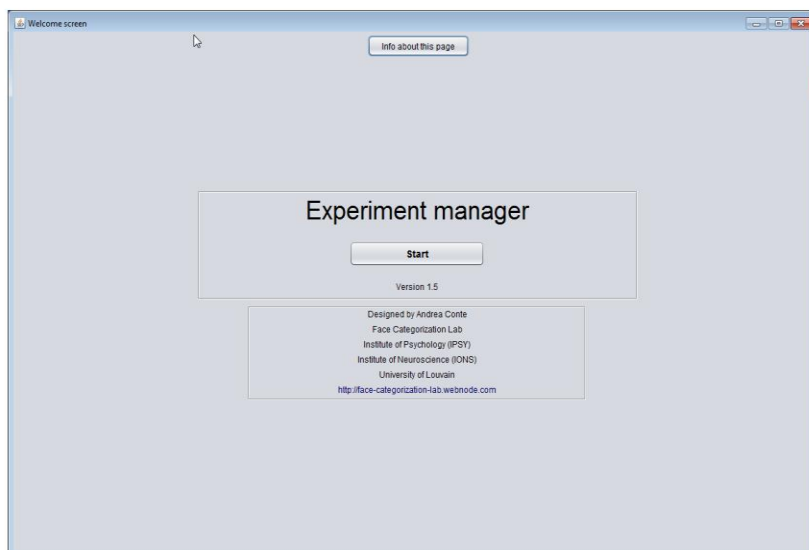
Start it!

After java runtime installed, start the application by clicking **Start.jar**.

If it starts for the first time, the application will make a **screen frequency test** in full screen mode.

Welcome Screen

After the screen refresh rate test, you will see the welcome screen:



Profile selection/definition

Profile Selection: After clicking **start**, you will see the selection profile screen:

Profile:

Back to welcome screen Info about this page

Select profile

Log in (or double click a profile in table to log in)

Create/Update profiles

Profile name	Profile ID	Owner	Information	Subjects visible to others	Programs visible to others
--------------	------------	-------	-------------	----------------------------	----------------------------

Profile definition: Click “**Create/Update profiles**” button to go to the define profile screen:

Profile:

Back to profile selection Save modification Info about this page

Profile definition

Profile name: Andrea Password: **** ☒ Subjects visible to others

Profile owner: Repeat password: **** ☒ Programs visible to others

Information:

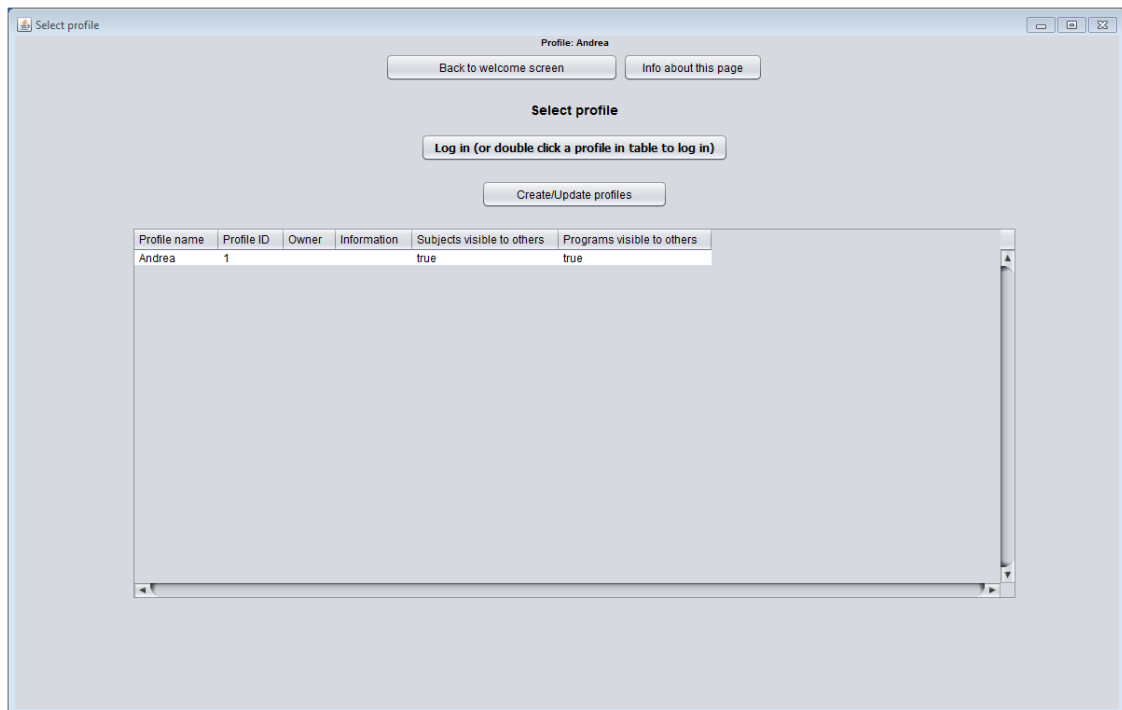
Profile name	Profile ID	Profile owner	Information	Subjects visible to others	Programs visible to others
Andrea	1			true	true

Enter a profile name and a password if needed (leave the password blank if not needed)

You can enter owner name and information if needed.

“Subjects visible to others” and *“Programs visible to others”* mean that others profiles can see your subjects and programs but cannot change them. The only thing they can do is to copy them in their profile. Once copied, they can make changes to the copied subject or program.

Profile Selection: Go back to profile selection, you will see the created profile:



Main page: Select the profile by double clicking or by clicking the Log in button to go to main page:



The 'Save modification' button

When a change is made, modification is applied automatically in the application but are not saved in the database. Click Save modification to save them definitively in the database file. The database file is called 'Profiles.dbo' and is found in ApplicationData/Database/ directory. In any case of unexpected error, you can lose the changes you made if you did not save the modification in the database file clicking that button.

Design experiments (Step 1)

Program definition

➔ Click the button **Program definitions**

Program definition

Program Parameters

Profile: Andrea

Back to main menu Save modification Info about this page

Program definition

Program name Resources main directory Change directory

Profile: Andrea Task name

ID in profile Parameters definitions file Select definitions file

Program Information

Select program and go to experiments definitions (or double click a program)

☐ Show visible programs from all profiles

Program name	Profile	ID in profile	Main resources directory	Task java class name	Task revision	Parameters file revision	Parameters definitions file	Creation date	Information
--------------	---------	---------------	--------------------------	----------------------	---------------	--------------------------	-----------------------------	---------------	-------------

Here you define the program:

Program name: Just give any name you want.

Profile: Do not need to fill in. Once the program is created, your profile name will appear here

ID in Profile: Do not need to fill in. Once the program is created, the program ID will appear here

Program information: Put whatever information you need. You can let it empty.

Resource main directory: It is the directory where you will put all your images. If you do not change directory, by default it will be “./ApplicationData/Resources” directory.

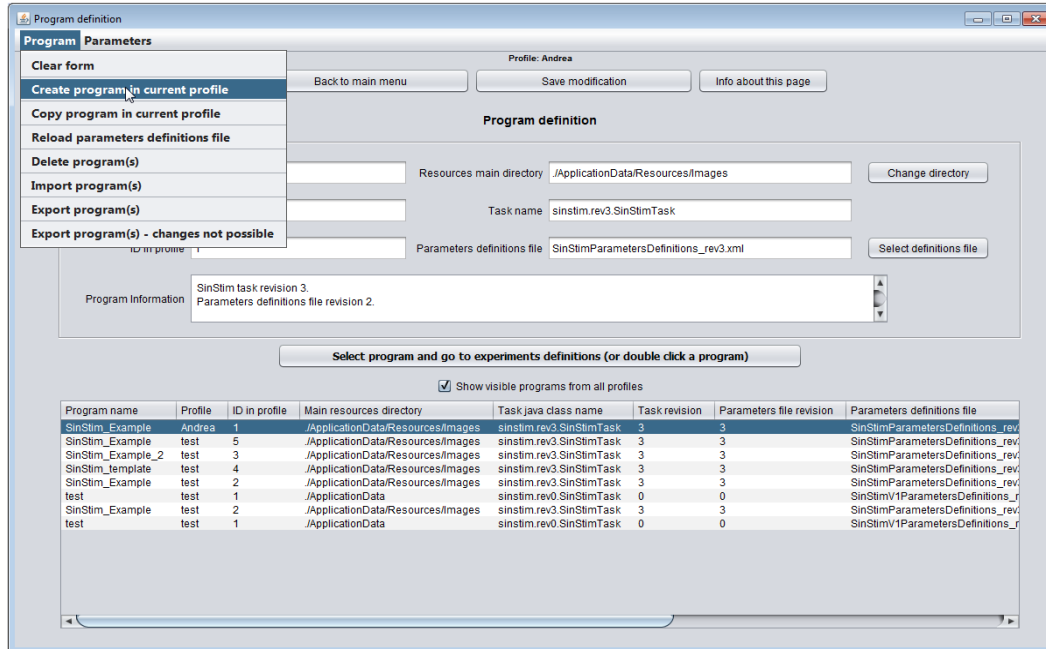
Task name: it is the name of the task coded in Java. For programmers, the name is actually a java class name. You have to put all package information first, following by the class name. Example: ‘sinstim.rev3.SinStimTask’ means class ‘SinStimTask’ in sub-package ‘rev3’ in package ‘sinstim’.

Parameter definitions file: it is a xml file where the task parameters are defined.

For example the parameters file for the task sinstim.rev3.SinStimTask is SinStimParametersDefinitions_rev3.xml. After the parameters are loaded from that file, you can modify them in program parameters, experiment parameters and condition parameters. When the task is started, it will apply the program, experiment and condition parameters to every trial.

Create program:

If all necessary fields are filled in, click on the top-down menu called Program and click 'Create program in current profile':



The 'Program' menu

Clear form: clear all fields

Create program in current profile: Create the new program

Copy program in current profile: Copy a program from your profile or from another profile (for that, you have to check 'show visible programs from all profiles')

Reload parameters definitions file: needed in case some new parameters were added to the parameters file.

Delete program(s): Delete one or more selected programs

Import program(s): Import one or more programs from an exported file. You can import, from directory ApplicationData/ProgramImport, a template program called 'SinStim_template.dbo' or an example program called 'SinStim_example.dbo'

Export program(s): Export one or more programs to a file

Export program(s) - changes not possible: Export one or more programs to a file but when importation is done later, no changes are possible in the imported program. It is just in reading mode.

The 'Parameters' menu

Select it to change the selected program parameters.

The 'Show visible programs from all profiles

Select to show all programs from profile that authorized program to be visible to others

Experiment definition

➔ Select and double click a program or click the button 'Select program and go..'

Experiment definition
Experiment Parameters

Profile: Andrea

Back to program selection Save modification Info about this page

Experiment definition
SinStim_Example

ID in program Experiment name Experiment Information

Select experiment (or double click) and go to next steps (Conditions and trial definitions)

Change experiments order

Experiment name	ID in program	Blocks will be randomized	Creation time	Experiment Information
-----------------	---------------	---------------------------	---------------	------------------------

ID in Program: Do not need to fill in. Once the experiment is created, the ID will appear here

Experiment name: Just give any name you want.

Experiment information: Put whatever information you need. You can let it empty.

Create Experiment:

If all necessary fields are filled in, click on the top-down menu called Experiment and click 'Create experiment':

Experiment definition
Experiment Parameters

Profile: Andrea

Back to program selection Save modification Info about this page

Experiment definition
SinStim_Example

Experiment name Experiment1 Experiment Information

Select experiment (or double click) and go to next steps (Conditions and trial definitions)

Change experiments order

Experiment name	ID in program	Blocks will be randomized	Creation time	Experiment Information
Experiment1	5	false	Thu Apr 09 17:12:47 CEST 2015	

The 'Experiment' menu

Clear form: clear all fields

Create experiment: Create the new experiment

Copy experiment: Copy an experiment

Delete experiment(s): Delete one or more selected experiments

Import experiment(s): Import one or more experiments in current program from an exported experiment. The parameters definition file used for the imported experiment must be the same as the current program parameters definition file.

Export experiment(s): Export one or more experiment to a file

The 'Parameters' menu

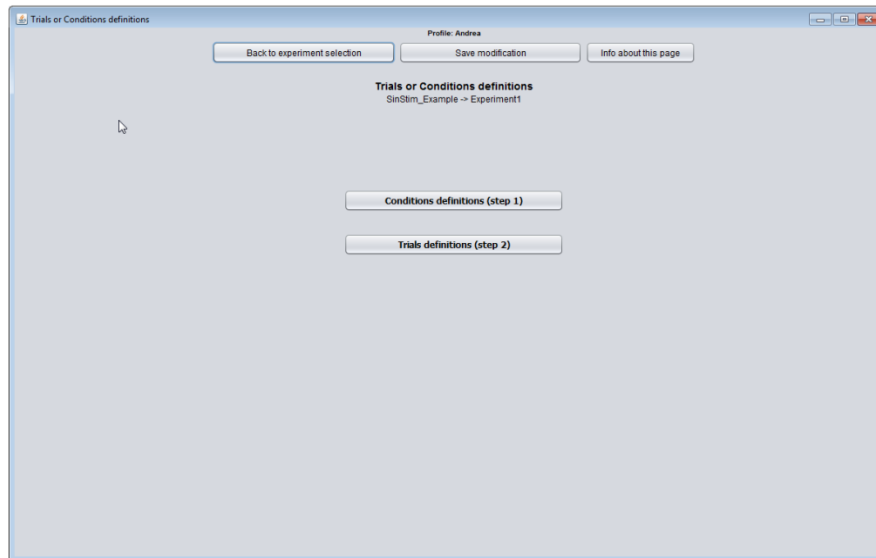
Select it to change the selected experiment parameters or the current program parameters where the experiment is currently defined.

The 'Change experiments order' button

If needed, you can change the order in which the experiments are shown in the table. But there is no other meaning of changing order than just a table view purpose.

Condition Definition

➔ Select and double click a experiment or click the button 'Select experiment and go..'



➔ Click Conditions definitions (step 1)

A screenshot of a software window titled 'Conditions definition'. The window has a light blue header bar with a profile icon and the name 'Andrea'. Below the header, there are three buttons: 'Back to conditions or trials definitions ...', 'Save modification', and 'Info about this page'. The main content area is light gray and contains the text 'Conditions definition' and 'SinStim_Example -> Experiment1'. There are two input fields: 'ID in experiment' and 'Condition name'. To the right of these fields is a text area for 'Condition information'. Below these fields, there are two buttons: 'Change conditions order' and 'Filter by name'. At the bottom, there is a table with the following headers: 'Condition name', 'ID in experiment', 'Creation time', and 'Condition information'. The table is currently empty.

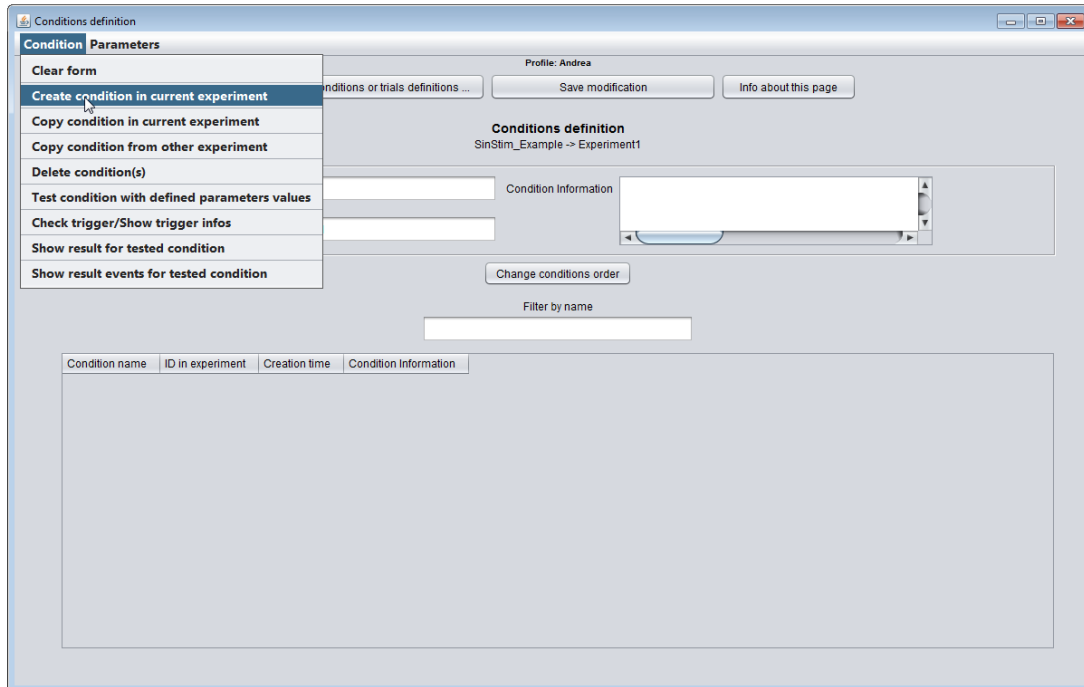
ID in Experiment: Do not need to fill in. Once the condition is created, the ID will appear here

Condition name: Just give any name you want.

Condition information: Put whatever information you need. You can let it empty.

Create Condition:

If all necessary fields are filled in, click on the top-down menu called Condition and click 'Create condition in current experiment':



The 'Condition' menu

Clear form: clear all fields

Create condition in current experiment: Create the new condition in the current experiment

Copy condition in current experiment: Copy the selected condition in the current experiment

Copy condition from other experiment: Copy a condition that is found in another experiment from the same program

Delete condition(s): Delete one or more selected conditions

Test condition...: Test a condition to be sure it works with all the parameters defined.

Check triggers/Show triggers information: Show all triggers that will be used information and verify if there are similar trigger numbers when the condition will be used in a trial.

Show results for tested condition: Show results in a table (example: the response time of the subject when it sees a stimuli or a fixation cross change)

Show result events for tested condition: Show events in a table. An event is something that is thrown and recorded in the application at a certain time. Example of events: An image is shown, a stimulation is starting, etc.)

The 'Parameters' menu

Select it to change the selected condition parameters, or the current experiment parameters where the condition is defined or the current program parameters where the experiment is defined.

The 'Change condition order' button

If needed, you can change the order in which the conditions are shown in the table. But there is no other meaning of changing order than just a table view purpose.

Filter by name

In case you have many conditions and you cannot find the one you need to modify

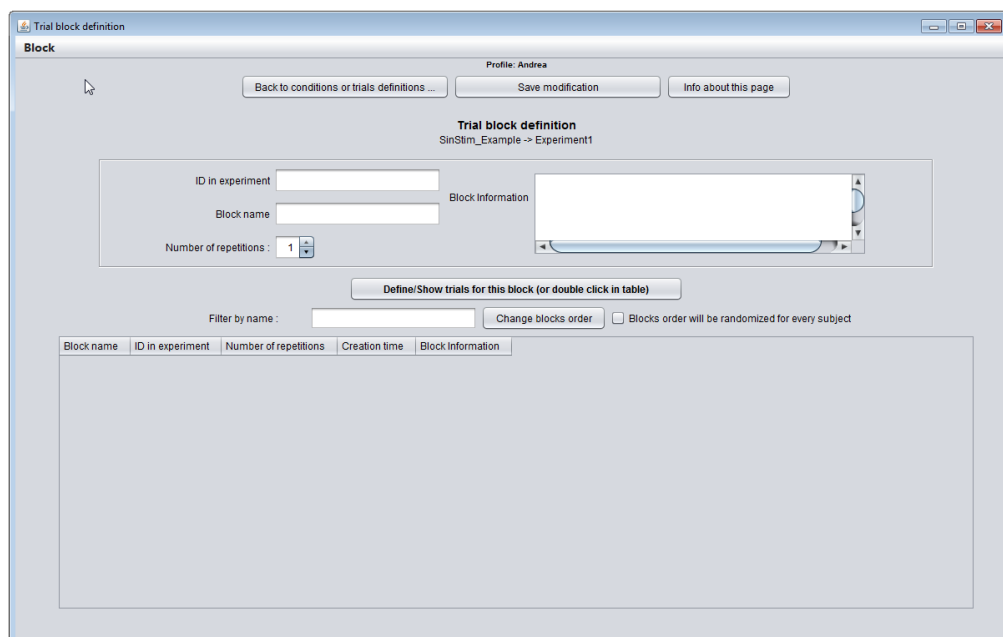
Trial Block Definition

(A block is a set of trials)

- ➔ In experiment definition, select and double click an experiment or click the button 'Select experiment and go..'



- ➔ Click Trials definitions (step 1), you will first have to define a trial block



ID in Experiment: Do not need to fill in. Once the trial block is created, the ID will appear here

Block name: Just give any name you want.

Block information: Put whatever information you need. You can let it empty.

Number of repetitions: You can ask to repeat that block a number of times with the same trials defined in it.

Create Block:

If all necessary fields are filled in, click on the top-down menu called Condition and click 'Create condition in current experiment':

Trial block definition

Profile: Andrea

Block menu:

- Clear form
- Create block
- Copy block
- Copy block from other experiment
- Delete block(s)

Experiment: 2

Block name: Block1

Number of repetitions: 2

Block Information

Buttons: Back to conditions or trials definitions ..., Save modification, Info about this page

Define/Show trials for this block (or double click in table)

Filter by name:

Change blocks order ☐ Blocks order will be randomized for every subject

Block name	ID in experiment	Number of repetitions	Creation time	Block information
Block1	2	2	Mon Apr 13 15:33:32 CEST 2015	
Block1	2	2	Mon Apr 13 15:33:32 CEST 2015	

The 'Block menu

Clear form: clear all fields

Create block: Create the new block in the current experiment

Copy block: Copy the selected block in the current experiment

Copy block from other experiment: Copy a block that is found in another experiment from the same program

Delete block(s): Delete one or more selected blocks

The 'Change blocks order' button

If needed, you can change the order in which the blocks will be shown to the subject.

Blocks order will be randomized for every subject

If selected, the order will be randomized for every subject whatever order you put in 'change blocks order'

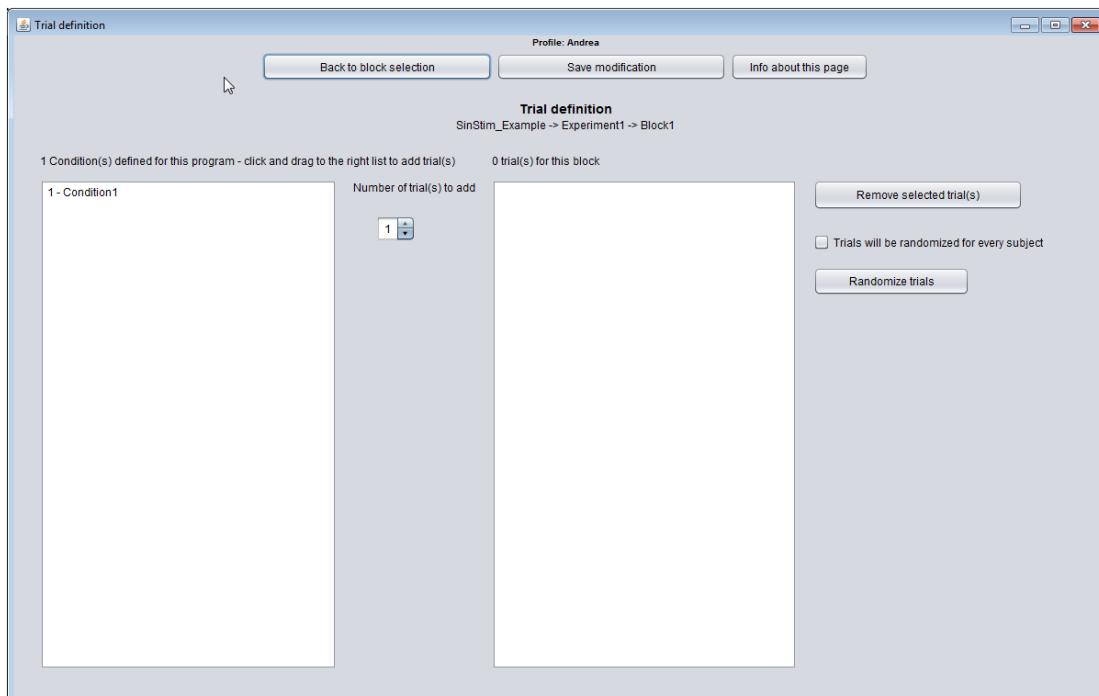
Filter by name

In case you have many blocks and you cannot find the one you need to modify

Trial Definition

Here you will defined the trials that compose the block

➔ Click 'Define/Show trials for this block...' or double click a block

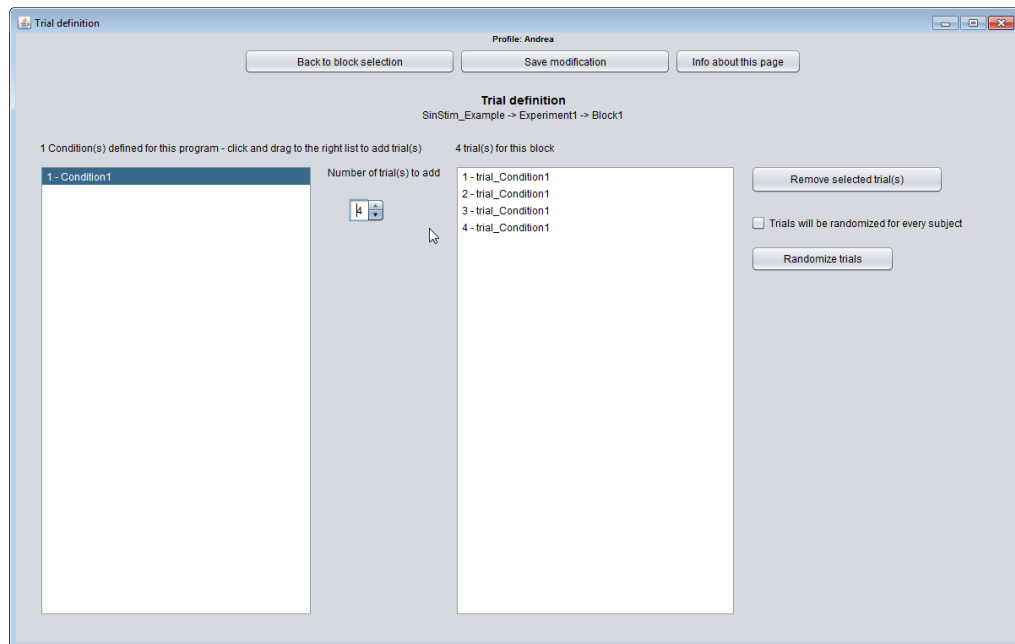


Create trials

On the left part, you have the list of conditions you have just defined.

Modify if needed 'Number of trials to add'.

Select one or more conditions from the left list. Drag and drop them from left list to right list. You will see the number of trials chosen be created in the right list for each condition selected.



Remove trials

To remove trials, just select them in the right list and click 'Remove selected trial(s)'

Change trials order

Select one or more trials and drag and drop them to another location in the list.

The order will be the same for all subjects using this block.

Randomize trials order

Just click the button 'randomize trials' to randomize them.

The random will then be the same for all subject using that block.

Randomize trials automatically for each subject

Select the check box 'Trials will be randomized for every subject'.

By selecting this option, the trials will be randomized every time a new subject will use that block whatever order you chose.

Subject definition

➔ On main page, click 'Subject definitions'

The screenshot shows a web application window titled "Subject definition". At the top, there's a "Subject" tab and a "Profile: Andrea" label. Below this are three buttons: "Back to main menu", "Save modification", and "Info about this page". The main section is titled "Subject definition" and contains a form with the following fields:

- Profile: Andrea (text input)
- Subject ID in profile: (text input)
- Other info 1: (text input)
- First name: (text input)
- Last name: (text input)
- Other info 2: (text input)
- Laterality: (dropdown menu)
- Sex: (dropdown menu)
- Other info 3: (text input)
- Birth date day/month/year ex:05/12/1998: (text input)

Below the form, there's a checkbox "Show visible subjects from all profiles" and two filter inputs: "Filter by first name" and "Filter by last name". At the bottom, there's a table with the following columns: First name, Last name, Profile, Subject ID in profile, Birth date, Sex, Laterality, Creation time, Other info 1, Other info 2, and Other info 3. The table is currently empty.

Fields

The minimum field needed is a first name or a last name.
You can put whatever information you need in other info 1, 2, 3

Subject Menu

The screenshot shows the same "Subject definition" window, but with the "Subject" menu open. The menu options are:

- Clear form
- Create subject
- Delete subject
- Import subject(s) (results is imported too)
- Export subject(s) (results is exported too)

The form fields are now populated with the following values:

- Profile: Andrea
- Subject ID in profile: 1
- Other info 1: (empty)
- First name: Andrea
- Last name: (empty)
- Other info 2: (empty)
- Laterality: (dropdown menu)
- Sex: (dropdown menu)
- Other info 3: (empty)
- Birth date day/month/year ex:05/12/1998: (empty)

The table at the bottom now contains one row of data:

First name	Last name	Profile	Subject ID in profile	Birth date	Sex	Laterality	Creation time	Other info 1	Other info 2	Other info 3
Andrea		Andrea	1				Mon Apr 13 16:48:18 CEST 2015			

Clear form: clear all fields

Create subject: Create a new subject from filling the fields

Delete subject(s): Delete one or more selected subjects

Import subjects(s): Import subjects and theirs results from an exported file.

Export subject(s): Export one or more subjects with theirs results to a file

The 'Show visible subjects from all profiles

Select to show all subjects from profile that authorized subjects to be visible to others

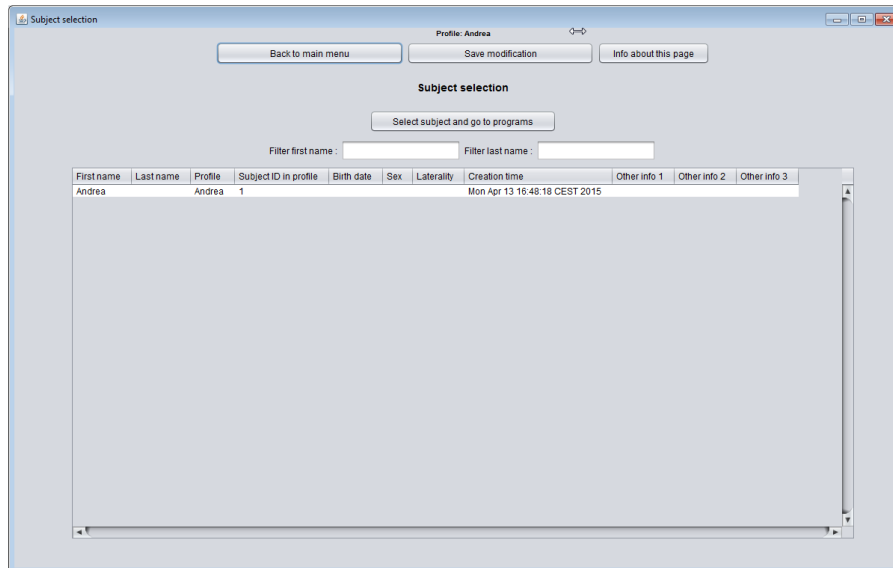
Filter by name or by last name

In case you have many subjects shown in the table and you cannot find the one you need to
modify

Run experiments (Step 2)

Subject selection

➔ On main page, click Run program, the subject selection page will appear

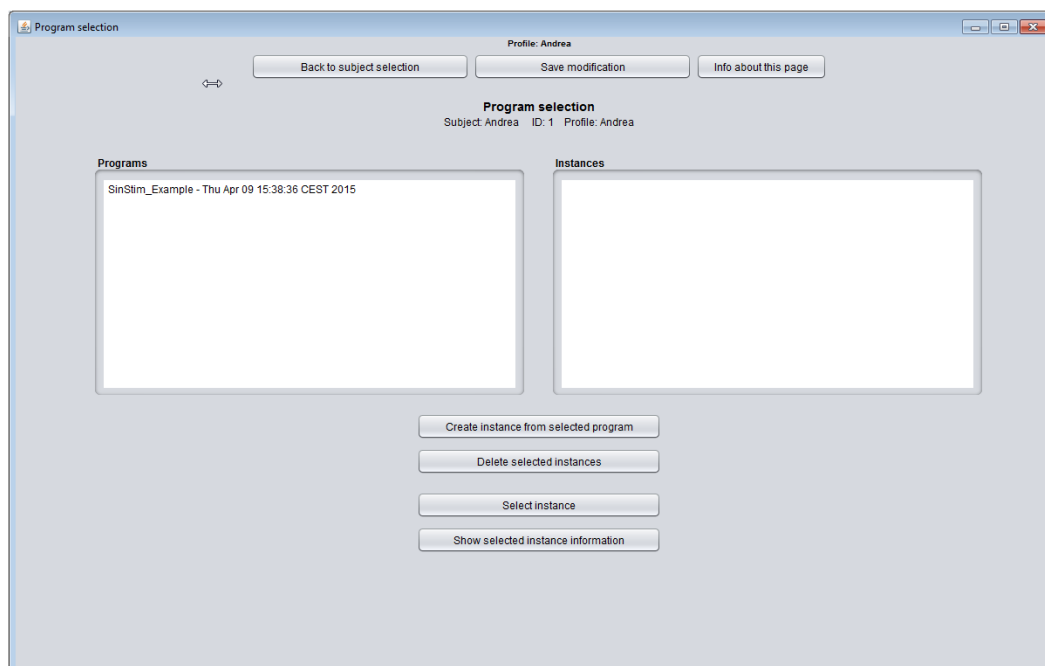


The screenshot shows a web application window titled "Subject selection". At the top, there is a "Profile: Andrea" label and a "Back to main menu" button. Below this, there are "Save modification" and "Info about this page" buttons. The main heading is "Subject selection", followed by a "Select subject and go to programs" button. Below this, there are two input fields for "Filter first name" and "Filter last name". A table displays the following data:

First name	Last name	Profile	Subject ID in profile	Birth date	Sex	Laterality	Creation time	Other info 1	Other info 2	Other info 3
Andrea		Andrea	1				Mon Apr 13 16:48:18 CEST 2015			

Program selection

➔ Double click a subject or press 'Select subject...' to go to program selection page



The screenshot shows a web application window titled "Program selection". At the top, there is a "Profile: Andrea" label and a "Back to subject selection" button. Below this, there are "Save modification" and "Info about this page" buttons. The main heading is "Program selection", followed by "Subject: Andrea ID: 1 Profile: Andrea". Below this, there are two panels: "Programs" and "Instances". The "Programs" panel contains a list item: "SinStim_Example - Thu Apr 09 15:38:36 CEST 2015". The "Instances" panel is empty. Below these panels, there are four buttons: "Create instance from selected program", "Delete selected instances", "Select instance", and "Show selected instance information".

Program instance

A program instance is a copy of a program, kind of photograph that you cannot modify once created.

You need program instance to launch a task.

Create program instance

Select on the left list an existing program, if some instances of that program already exist, you will see them on the right list.

‘Create instance from selected program’ will create a new instance of that program. You can give it any name you want. Program instances are created for all subjects; you don’t need to create a new one every time you select a subject.

Delete selected instances

Delete selected instances linked to that program.

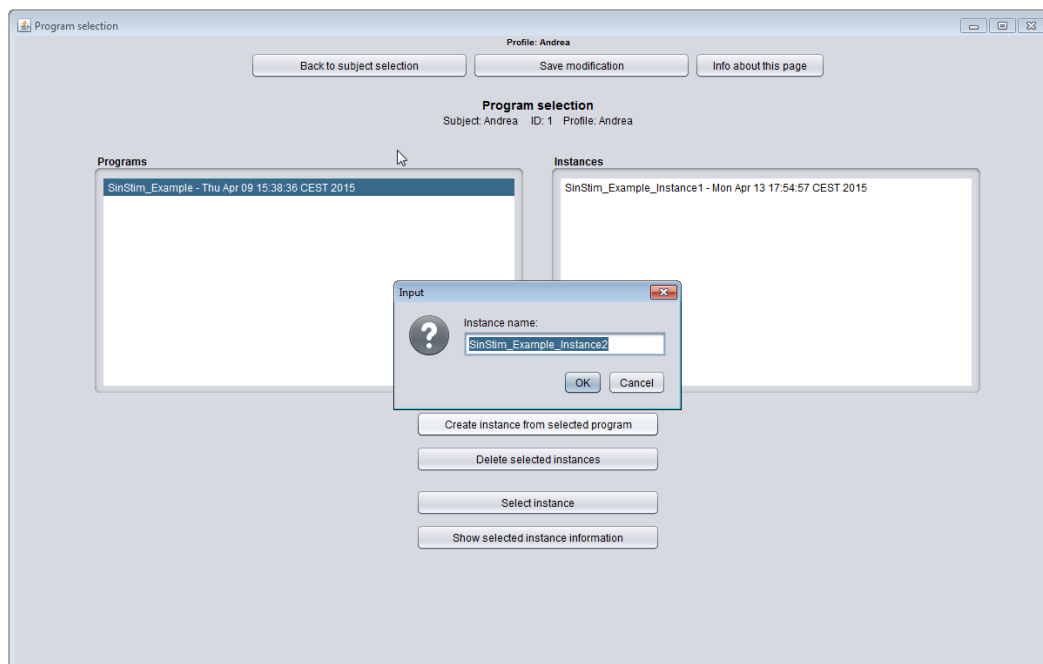
If an instance is deleted; it will not delete the subject result linked to that instance. You will still be able to see that subject results.

Select instance

Select an instance and go to experiment selection

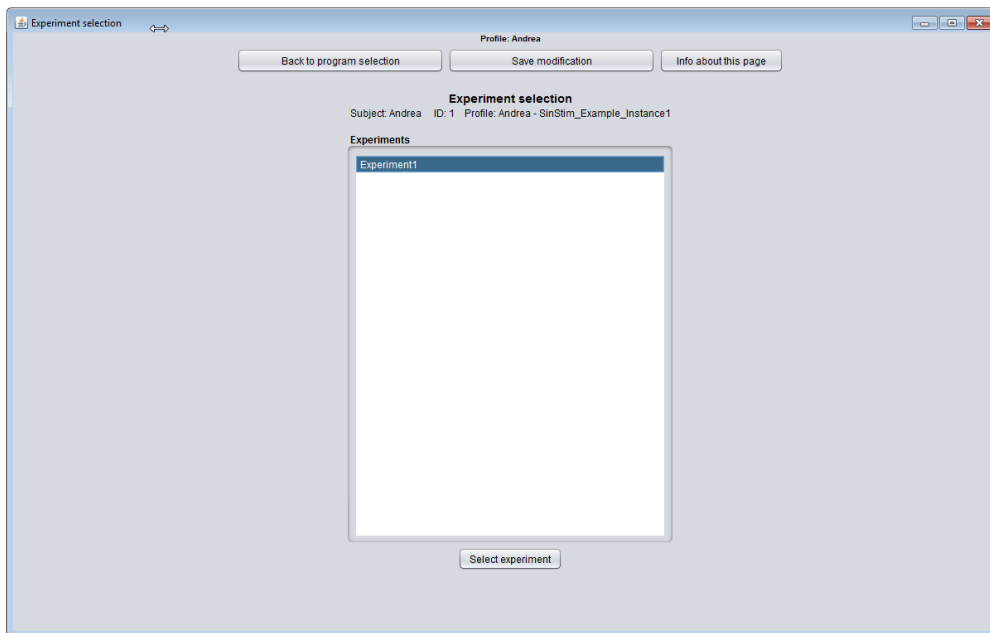
Show selected instance information

Show information relative to that program instance. It means all the information related to the program linked to that instance including the parameters values it had when the instance was created.



Experiment selection

➔ Click Select an instance to go to experiment selection



➔ Select an experiment in the list and double-click or click button 'Select experiment'

Run the task

Many different options are available to launch the task

